INCH-POUND
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PERFORMANCE SPECIFICATION

PUMPING ASSEMBLIES, WHEEL-MOUNTED, DIESEL ENGINE DRIVEN, BULK TRANSFER, FUEL AND POTABLE WATER PUMPING SERVICE

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This purchase description covers wheel-mounted, diesel-engine-driven, self-priming, four-inch and six-inch hose connection pumps for fuel or potable water service, hereinafter referred to as "pump assembly".

1.2 Classification. The pump assemblies will be of the following types as specified (see 6.2).

Type I - Pump Assembly, Fuel

Size 1 - 350 gallons per minute (GPM) Capacity at 275 Feet Total Head

Size 2 - 600 GPM Capacity at 350 Feet Total Head
(1250 GPM Capacity at 180 feet total head)

Type II - Pump Assembly, Water

Size 1 - 350 GPM Capacity at 275 Feet Total Head

Size 2 - 600 GPM Capacity at 350 Feet Total Head

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AFLP-3747	- Guide Specifications for Aviation Turbine Fuels.
AMOVP-1	- Road Movements and Movement Control.
STANAG 1135	- Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations.
STANAG 4074	- Auxiliary Power Unit Connections for Starting Tactical Land Vehicles.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52624	- Antifreeze, Multi Engine Type
A-A-59326	- Coupling Halves, Quick-Disconnect, Cam-Locking Type.
A-A-59592	- Can, Fuel, Military: 20-Liter Capacity.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-DTL-64159	- Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-DTL-83133	- Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37).
MIL-PRF-2104	- Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service.
MIL-PRF-10924	- Grease, Automotive and Artillery.
MIL-PRF-21260	- Lubricating Oil, Internal Combustion Engine, Preservative Break-In.
MIL-PRF-32143	- Batteries, Storage: Automotive, Valve Regulated Lead Acid
MIL-PRF-32349	- Container, Cargo Triple Container (TRICON) (Without Cabinets, Drawers, or Shelves).
MIL-PRF-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	- Military Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Lifting and Tiedown Provisions.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-464	- Electromagnetic Environmental Effects Requirements for Systems.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-1179	- Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles.
MIL-STD-1366	- Transportability Criteria.
MIL-STD-1472	- Human Engineering.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

CFR Title 40	- Protection of Environment.

(Copies of this document are available online at www.ecfr.gov.)

US ARMY PAMPHLETS

DA Pam 611-21	- Military Occupational Classification and Structure.

(Copies of this document are available from usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil.)

US ARMY TECHNICAL MANUALS

TM 9-2320-211-10	- Truck, 5-Ton, 6x6, M39 Series (Multifuel), Operators Manual for.
TM 9-2320-260-10	- Truck, 5-Ton, 6x6, M809 Series (Diesel), Operators Manual for.
TM 9-2320-272-10	- Truck, 5-Ton, 6x6, M939, M939A1, and M939A2 Series Trucks (Diesel), Operators Manual for.
TM 9-2320-366-10	- M1083 Series, 5 Ton, 6x6, Medium Tactical Vehicles (MTV), Operators Manual for.
TM 10-3930-643-10	- Truck, Forklift, DED Pneumatic Tire, 10,000 lb. Capacity,

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Rough Terrain, Articulated Frame Steer, (M10A), Operators Manual for.

TM 10-3930-673-10 - All Terrain Lifter Army System (ATLAS), Clean Burn Diesel, 10,000 lb. Capacity, Operators Manual for.

(Copies of these documents are available online at https://armypubs.army.mil.)

DRAWINGS

11674728 - Vehicle Receptacle Assembly, NATO Intervehicle Power.
11682336 - Cable and Plug Assembly, Intervehicle Power.
13219E0462 - Ground Rod

(Copies of these documents are available from usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
ASME B1.20.1 - Pipe Threads, General Purpose (Inch).
ASME B1.20.3 - Dryseal Pipe Threads (Inch) Revision and Redesignation of B2.2-1968.
ASME BPVC IX - Boiler and Pressure Vessel Code (BPVC). Section IX Qualification Standard for Welding, Brazing, and Fuzing Procedures

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D975 - Standard Specification for Diesel Fuel.
ASTM D1655 - Standard Specification for Aviation Turbine Fuels.

(Copies of these documents are available online at www.astm.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M - Structural Welding Code - Steel.
AWS D1.2/D1.2M - Structural Welding Code - Aluminum.
AWS D1.3/D1.3M - Structural Welding Code - Sheet Steel.

(Copies of these documents are available online at www.aws.org.)

HYDRAULIC INSTITUTE (HI)

ANSI/HI 14.6 - Rotodynamic Pumps for Hydraulic Performance Acceptance Tests

(Copies of this document are available online at www.pumps.org.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

(Copies of this document are available online at www.nema.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement
SAE J514 - Hydraulic Tube Fittings (DoD Adopted)
SAE J614 - Engine and Transmission Dipstick Marking (DoD Adopted)
SAE J847 - Trailer Tow Bar Eye and Pintle Hook/Coupler Performance
SAE J2360 - Automotive Gear Lubricants for Commercial and Military Use

(Copies of these documents are available online at www.sae.org.)

TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Year Book

(Copies of this document are available online at www.us-tra.org).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with (IAW) 4.1.1.

3.2 Materials. The contractor shall verify that the materials used in the pump assembly are as identified and shall meet all of the operational and environmental requirements specified. The components incorporated into the pump assembly shall be newly fabricated from recovered materials (see 6.6.1) to the maximum extent practicable, provided the components meet all other requirements of this document. Type I pump assemblies shall be constructed of materials suitable for fuel pumping service. Type II pump assemblies shall be constructed of materials suitable for potable water pumping service. Lead shall not be used. The finished Type II pump assembly shall contain no materials or substances that might leak or disintegrate and cause the pumped water to become non-potable. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that contact potable water shall conform to the applicable federal regulations for use with potable water.

3.2.1 Deterioration prevention and control. The pump assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against corrosion and deterioration during storage and operational conditions experienced.

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3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt, or remanufactured components shall not be incorporated into the pump assembly.

3.2.4 Usage of COTS items. Commercial-off-the-shelf (COTS) items shall be used whenever available. COTS items shall not be modified unless absolutely necessary.

3.3 Operating requirements.

3.3.1 Pump assembly performance.

3.3.1.1 Hydrostatic pressure. The pump assembly shall be rated for a maximum operating pressure of 150 psig. The pump assembly, with the valves open and with couplers' dust covers installed (see 3.3.2.3), shall withstand an internal hydrostatic pressure of 225 pounds per square inch gage (psig) for not less than 5 minutes without rupture or distortion to any component, and without leakage through the assembly, fittings, piping, or joints.

3.3.1.2 Capacity. The pump assembly shall meet the rated flow capacities indicated below using water at standard conditions, as specified by the Hydraulic Institute ANSI/HI 14.6 Centrifugal Pumps and Tests. Rated flow rates are as follows:

a. Type I, Size 1	- 350 gpm with a total dynamic head of 275 feet.
b. Type I, Size 2	- 600 gpm with a total dynamic head of 350 feet, and 1250 gpm with a total dynamic head of 180 feet.
c. Type II, Size 1	- 350 gpm with a total dynamic head of 275 feet.
d. Type II, Size 2	- 600 gpm with a total dynamic head of 350 feet.

3.3.1.2.1 Tilting. The pump assembly shall perform at rated flow as specified in 3.3.1.2 when tilted a minimum of 15 degrees from horizontal in any direction.

3.3.1.3 Pump priming. The pump assembly shall prime automatically, after initial filling of the case, and deliver not less than the rated flow (see 3.3.1.2) in not greater than 2 minutes, when operating at a static suction lift equivalent to 10 ft at sea level and atmospheric conditions of 29.92 inches of mercury, using water with a specific weight of 62.3 pounds per cubic foot (lbs/ft^3) at 68 degrees Fahrenheit (°F).

3.3.1.4 Slope capability. The pump assembly shall be capable of negotiating a minimum of 20 percent (%) side slopes; and approaching, negotiating, and departing a ramp of a minimum 30% slope without ground contact of any component (except the tires), and without contact of the tires with any other part of the assembly.

3.3.1.5 Mobility. The pump assembly, when towed by any designated prime mover (see 3.3.1.5.1), shall have mobility not less than that described below without malfunction, permanent deformation, or damage.

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3.3.1.5.1 Prime movers. The pump assembly shall be capable of being towed by standard US Army 5-ton Series Tactical Cargo Trucks of the types and series described below:

a. M39 Series, TM 9-2320-211-10.
b. M809 Series, TM 9-2320-260-10.
c. M939 Series, TM 9-2320-272-10.
d. M939A1 Series, TM 9-2320-272-10.
e. M939A2 Series, TM 9-2320-272-10.
f. M1083 Series "FMTV", TM 9-2320-366-10.

3.3.1.5.2 Mobility.

a. Improved (paved) roads at an average speed of 20 miles per hour (mph).
b. Unimproved (unpaved) roads at an average speed of 10 mph.
c. Over cross country terrain at an average speed of 8 mph.

3.3.1.6 Drop. The pump assembly shall be capable of withstanding drops of 26 inches, measured from the lunette to the ground, onto its lunette and frame, about its axle (see 3.3.2.9) without permanent damage, deformation, or any degradation in performance.

3.3.1.7 Fording capability. The assembly shall be capable of fording in water to a depth of not less than 30 inches, without modification or kits, without damage or degradation in performance.

3.3.2 Pump assembly characteristics and components. All threaded parts shall conform to ASME B1.1, ASME B1.20.1, or ASME B1.20.3, as applicable.

3.3.2.1 Pump housing. The pump housing shall be provided with a minimum 1.5-inch priming port with locking cap, for filling the pump housing without removal of the suction or discharge connections. The pump housing shall be provided with a drain port and manual valve for draining the case without tilting, and capable of being fully drained in five (5) minutes or less. The valve discharge shall be routed for easy accessibility and to allow for collection of the discharge and prevent spillage into the environment (see 3.3.2.8.1 and 3.5.3). The priming port shall be equipped with a male, quick-disconnect, coupling and captive dust cap with gasket IAW A-A-59326.

3.3.2.2 Piping and fittings. All piping and fittings shall be mounted securely and shall withstand an internal working pressure of 150% of the shutoff pressure or maximum operating pressure, whichever is higher, at rated speed without leaking or deformation. The discharge piping shall contain a vent valve, with 0.25-inch NPT thread IAW ASME B1.20.1, to remove air during priming. The vent shall be located at the highest point in the discharge piping. The vent valve discharge shall be routed for easy accessibility and to allow for collection of the discharge and prevent spillage into the environment (see 3.3.2.8.1 and 3.5.3).

3.3.2.3 Suction and discharge components. The pump assembly shall have the following suction and discharge components:

a. Discharge check valve, of a durable, light-weight, corrosion resistant, non-magnetic material, suitable to the application and compatible with discharge housing.

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b. Two four-inch valves, minimum size, shall be incorporated on the suction port, and on the discharge port, of the pump. The valves shall be supported by brackets connected to the assembly chassis, and shall be labeled to indicate the open-close direction. The valves shall be positioned to accept both hose and pipe.
c. The suction side of the pump shall terminate in a six-inch and a four-inch, female, quick-disconnect couplings with gaskets IAW A-A-59326, Class A, fitted with tethered dust plugs IAW A-A-59326/11, Class A.
d. The discharge side of the pump shall terminate in a six-inch and a four-inch, male, quick-disconnect couplings IAW A-A-59326, Class A, fitted with tethered dust caps and gaskets IAW A-A-59326/10, Class A.
e. A suction strainer assembly, consisting of a 0.25 inch mesh, in-line type, reinforced basket strainer of a durable, rigid, corrosion resistant material suitable to the application.

3.3.2.4 Engine. The engine shall be rated by the manufacturer for continuous power duty adequate to power the assembly to meet the performance requirements of 3.3.1.2. The engine shall meet the performance requirements herein with JP-8 as the primary fuel (see 3.4.3). Diesel engines shall have an EPA emission certification for non-road diesel engines as required by CFR Title 40 Parts 9, 86, and 89 at its power rating. The engine shall be equipped with, but not limited to, the following items:

a. A nominal 24-volt direct current (VDC) electric cranking system.
b. An electrical charging system.
c. A heavy duty, dry type, air cleaner with replaceable cartridge.
d. Fuel filters: primary, water-separating type, and secondary (final); each with provision for drainage into an external container (see 3.3.2.8.1).
e. Exhaust system with guard and rain cap.
f. Engine oil drain, easily accessible for maintenance and to minimize the potential for spillage to the environment (see 3.5.3).
g. Heavy duty cooling system capable of maintaining the manufacturers recommended operating temperature when performing at maximum rated capacity as specified herein. If liquid-cooled, the engine shall operate with antifreeze IAW A-A-52624; cooling system overflow reservoir shall be included for liquid-cooled engines, to prevent coolant discharge into the environment (see 3.5.3).
h. Engine rotating and moving parts guard, providing reliable, quick, and easy access for maintenance, and requiring no tools for removal or installation.
i. Oil dipstick conforming to SAE J614.

3.3.2.5 Electrical system. The starting, charging and lighting systems, and all electrical components of the assembly shall be nominal 24 VDC. A temperature-compensated, integral charging system, wiring, and battery(s) shall be supplied to provide starting, lighting, and additional electrical power as needed to (a) recharge the battery(s) in not more than 30 minutes of continuous operation after the system has been started three times at -25°F, and (b) start the system at -25°F three times within approximately a 1-hour period with a minimum of 15 minutes between starts after a 5 minute run period.

3.3.2.5.1 Batteries. All batteries shall be US Army approved IAW MIL-PRF-32143. Batteries shall be readily accessible for service and inspection and shall be of sufficient quantity

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and type to comply with the starting, lighting, normal and surge electrical loads, reserve electrical power capacity, and maintenance requirements.

3.3.2.5.2 Battery cables. The positive cable(s) shall have a non-conductive red sleeve at both ends. The negative cable(s) shall have a black non-conductive sleeve on both ends. A battery terminal non-conductive protective cover shall be provided on each battery terminal connector (red for positive and black for negative). The battery cables shall be marked to indicate polarity.

3.3.2.5.3 Battery box. A vented battery box shall be provided to retain/protect the batteries during operation and transport. The box shall allow for easy installation and removal of the batteries. The box shall protect the batteries from external environmental conditions and shall provide for venting of battery gases at or near the top of the box. The box shall be resistant to battery electrolyte. Means shall be provided to allow drainage out of the bottom of the box, regardless of incline.

3.3.2.6 Control and Instrument panels. A control panel and an instrument panel shall be provided for the pump assembly. The control and instrument panels shall be covered by a latching door(s) with automatically engaging hold-open devices. The panels shall each contain a switch that automatically turns off the lights when the panel door is closed. All panel face components shall be labeled and legible by the operator.

3.3.2.6.1 Control panel. The control panel shall be of modular design and meet the requirements for NEMA 250 Type 4 X enclosures. The control panel shall be replaceable in under 10 minutes using no special tools (see 6.6.5). The control panel shall include the following as a minimum:

a. Power ON/OFF toggle circuit breaker and indicator light.
b. Assembly OFF/RUN/START toggle switch and indicator light.
c. Cold start assist power toggle switch (momentary), if applicable, and indicator light.
d. Mode selector switch (3 position; MANUAL/SPEED/PRESSURE).
e. RAISE/HOLD/LOWER toggle switch (return to center).
f. Suction pressure limit override toggle switch and indicator light.
g. Engine shutdown indicator(s).
h. Battle short bypass toggle switch and indicator light.
i. Voltmeter.
j. Ammeter.
k. EMERGENCY STOP pushbutton, readily accessible in case of emergency, even if control panel door is closed.

3.3.2.6.1.1 Control panel components. The control panel face components in 3.3.2.6.1 shall be sealed against the environmental effects of 3.6.3 (Sand) through 3.6.8 (Salt fog). Components located in the control panel shall only be electrical components. Discrete visual indicators shall be of the light emitting diode (LED) type in lieu of incandescent bulbs. All switching components shall be hermetically sealed.

3.3.2.6.1.2 Engine/pump protection. The engine and pump of the assembly shall be provided with automatic protections resulting in engine shutdown. A manual reset shall be required following an automatic shutdown; toggling the assembly OFF/RUN toggle switch is

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acceptable. All alarm and shutdown circuits shall be designed fail-safe; a shutdown condition, sensor failure, broken wire, blown fuse, etc. shall result in an engine shutdown. Engine/pump protections shall include the following shutdown conditions as a minimum:

a. Engine over speed (exceeding 110% of maximum engine rated speed).
b. High engine temperature.
c. High fuel tank temperature (exceeding 175 degrees F).
d. Low engine oil pressure.
e. Excessive discharge pressure (exceeding 110% of rated discharge pressure, see 3.3.1.2)

3.3.2.6.1.3 Shutdown annunciation. Shutdown annunciation shall be provided for the engine/pump protections of 3.3.2.6.1.2. A unique indication shall be provided for each shutdown condition and each sensor failure. A diagnostic or start-up check shall be provided allowing the operator to verify that the shutdown indicator(s) is(are) functional.

3.3.2.6.1.4 Emergency shutdown capability. The pump assembly shall be equipped with a reliable, fail-safe, remote emergency shutdown capability. Operator shall have the ability to shut down the engine/pump beyond a 15 foot perimeter of the pump assembly. The remote emergency shutdown shall be capable of secure mounting on the ground rod (see 3.5.5.2), to be readily available for use and not resting on the ground surface.

3.3.2.6.2 Instrument panel. Components located in the instrument panel shall only be mechanical type components. All display gauges shall be analog gauges even where electrical signals to the panel are allowed, and liquid process lines shall be physically separated from electrical components in the instrument panel in case of liquid leaks. The instrument panel shall include the following as a minimum:

a. Suction pressure gauge of the Bourdon type, compound with a minimum range of 30 inches Hg (mercury) vacuum - 0 psig - 200 psig.
b. Discharge pressure gauge of the Bourdon type, with a minimum range of 0 - 200 psig.
c. Analog tachometer-hour meter. May be an electrical signal converted at the panel to analog display.
d. Analog engine oil pressure gauge. May be an electrical signal converted at the panel to analog display.
e. Analog engine temperature gauge. May be an electrical signal converted at the panel to analog display.
f. Air filter restriction indicator.
g. Manual throttle (see 3.3.2.7.1).

3.3.2.6.3 Pressure lines. Any process lines that provide process fluid to the instrument panel shall be minimum 0.25 inch stainless steel tubing with short runs of stainless steel braided hoses in areas that require movement (vibration isolation). Readily accessible drainage cocks shall be provided as required and lines shall be mounted to provide complete drainage for storage and transport. The discharge shall be routed for easy accessibility and to allow for collection of the discharge and prevent spillage into the environment (see 3.3.2.8.1 and 3.5.3).

3.3.2.7 Automatic pump controller. The assembly shall include controls for placing the pumping assembly in three different modes of operation: manual, automatic pressure control, and automatic speed control. In either of the automatic controller modes, the assembly shall be

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equipped to reduce engine speed to limit high discharge pressure or low suction pressure. The automatic controller shall sense the discharge pressure and begin to reduce engine speed at 150 psig. If discharge pressure continues to increase, the controller shall continue to reduce engine speed until the idle setting is reached. To prevent pump cavitation and damage to the pump and engine, the automatic controller shall also sense the pressure on the suction side of the pump. If suction pressure falls to 10 psig, the controller shall begin reducing engine speed. If suction pressure continues to decrease, the controller shall continue to reduce engine speed until the idle setting is reached. The controller shall be capable of being turned off in the manual mode so that engine speed is regulated via the manual throttle.

3.3.2.7.1 Manual mode. When operating in manual mode, the automatic controller features of 3.3.2.7 shall be overridden by the manual throttle. The manual throttle shall control engine speed via manual means acting directly on the engine original equipment manufacturer (OEM) component(s) (preferably acting on mechanical governor).

3.3.2.7.2 Automatic speed control mode. When operating in automatic speed control mode, the RAISE/HOLD/LOWER toggle switch of 3.3.2.6.1 shall be used to adjust engine speed from the minimum (idle) to the maximum operating speed. The automatic pump controller shall control the engine speed to the selected setting. The automatic pump controller of 3.3.2.7 shall override the selected setting if high discharge pressure and/or low suction pressure limits are reached.

3.3.2.7.3 Automatic pressure control mode. When operating in automatic pressure control mode, the RAISE/HOLD/LOWER toggle switch of 3.3.2.6.1 shall be used to adjust discharge pressure from the minimum to the maximum operating discharge pressure. The automatic pump controller shall control the engine speed to meet the selected discharge pressure. The automatic pump controller of 3.3.2.7 shall override the selected setting if high discharge pressure and/or low suction pressure limits are reached.

3.3.2.7.4 Suction pressure limit override. The assembly shall be capable of operating in either of the automatic controller modes with the low suction pressure limit enabled or disabled. The low suction pressure limit shall be enabled or disabled via the suction pressure limit override toggle switch located on the control panel of 3.3.2.6.1.

3.3.2.7.5 Battle Short bypass. The assembly shall include a battle short bypass. This feature shall bypass any engine shutdown controls that would normally remain operational when the assembly is in manual mode, including those of 3.3.2.6.1.2, for operation of the assembly with a damaged control panel in a battlefield environment. This feature shall have an indicator, with label, on the face of the control panel to signal the operator(s) that the bypass is engaged.

3.3.2.8 Fuel system. A fuel tank, sufficient for not less than 8 hours of continuous operation at the engine manufacturer's maximum rated load, shall be furnished. A fuel gauge shall be provided, graduated in ¼-tank increments, minimum, from empty to full, and located on the fuel tank. The fuel tank shall include a hand-operated fuel shutoff valve with an arrow showing the directions of valve operation. The fuel tank pickup tube shall be from 0.50-inch to 1-inch above the bottom of the tank. The non-plastic fuel line shall be provided with a 3-way fuel selector valve and a 0.50-inch SAE J514 flared fitting for connection to an auxiliary fuel supply. A corrosion-resistant circular metal face-plate for the fuel selector valve shall be marked "OFF", "UNIT TANK" and AUXILIARY". No fuel spillage shall occur when the fuel tank is

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full and the assembly is operating IAW 3.3.1.2. The body of the fuel tank shall be fabricated from non-metallic material; corrosion-resistant metallic inserts may be used at connection points, interfaces, and ports. Surfaces shall be designed to shed water and moisture regardless of incline (see 3.6.7). The fuel tank shall be protected from damage by stones and other projectiles when in tow.

3.3.2.8.1 Fuel reservoir. One Military Fuel Can, 5 gallon (20 Liter) capacity, IAW A-A-59592 shall be included with each pump assembly. Mounting for the fuel reservoir on the pump assembly shall be provided; Bracket Assembly, A-A-52513, is provided as reference only. The fuel can shall be used to receive fuel discharge from the vent valve (see 3.3.2.2) and pressure lines (see 3.3.2.6.3) when operating a Type I (fuel) pump assembly, and for minor refueling of the fuel system (see 3.3.2.8). The fuel reservoir shall include an additional cap to be used when the pump assembly is in operation. The additional cap shall allow for venting of pressure, prevent dirt and contamination from entering the fuel reservoir, and have connections for the vent valve discharge routing (see 3.3.2.2) and pressure line drainage (see 3.3.2.6.3).

3.3.2.8.2 Refueling from pumped fluid. The Type I (fuel) pump assembly shall be equipped with the capability of refueling (filling of the fuel tank) from the pumped fluid while pumping; the Type II (water) pump assembly shall be prohibited from having this capability. A manual valve shall be included at the discharge of the pump, and a fail closed, hold to open, valve at the fuel tank. The design shall result in safe refueling under all operating conditions.

3.3.2.9 Chassis. The pump assembly components shall be mounted on a chassis having two single wheels, on an axle assembly attached to the underside of the assembly frame. The axle and chassis assembly shall have a rated capacity exceeding the maximum payload, but not less than 6000 lbs. The chassis shall be equipped with a minimum of two shock absorbers for stability. The axle assembly(s) may be of the torsion type in lieu of the two shock absorbers.

3.3.2.9.1 Tires and wheels. Tires and wheels shall be IAW the guidelines of the Tire and Rim Association Year Book and shall be of sufficient size and durability to support loads in off-road conditions. Tire pressure shall be indicated directly above the tires in 1-inch high letters.

3.3.2.9.2 Brakes. The pump assembly shall be equipped with parking brakes capable of keeping the assembly from moving down a 25% slope. The parking brake shall be manually operated with snap-over-center locking design. The force required to apply the parking brake shall not exceed 35 lbs. The parking brake handle shall be accessible to the operator. All cables and brake lines shall be routed to not interfere with the free movement of other components. The brake lines and all cables shall be securely clamped along the routing path and routed to avoid brush and debris damage.

3.3.2.9.3 Towing extension. The pump assembly shall have a towing eye (lunette) IAW SAE J847 with a nominal height of 26-inch when the unit is level and on a level surface. Weight on the towing eye with the fully-fueled assembly parked on level ground shall be not greater than 15% and not less than 10% of the gross weight of the assembly. When necessary, the assembly shall be equipped with retractable towing extensions capable of easy extension for towing and easy retraction for storage.

3.3.2.9.4 Stabilizing legs. The pump assembly shall be equipped with an adjustable, hand-crank style jack to permit operation of the towing extension by one person (see 3.5.2 and

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3.7). Hand-crank style stabilizing legs shall be provided to permit assembly operation without tipping over when not attached to the designated prime mover (see 3.3.1.5.1). The jack and stabilizing legs shall each be equipped with a sandshoe with a minimum area of 150 square inches (in^2). When not in use, the jack and stabilizing legs shall be positioned in such a way as to not affect the towing of the assembly. The jack and legs shall each have a minimum travel of 10-inches.

3.3.2.9.5 Lights, reflectors and connector. Pump assembly shall be equipped with reflex reflectors IAW MIL-STD-1179.

3.4 Interface requirements.

3.4.1 Transportability. The pump assembly shall be capable of being transported by military or commercial tractors, tractor/trailer combinations, trains (rail), marine vessels, and aircraft per MIL-STD-1366 and 3.4.1.6. The pump assembly shall be equipped with tiedown and slinging provisions. The pump assembly shall be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation.

3.4.1.1 Tiedown provisions. The pump assembly shall be equipped with tiedown provisions IAW MIL-STD-209. The tiedown provisions shall restrain the components without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.4.1.2 Slinging provisions. The pump assembly shall be equipped with slinging provisions IAW MIL-STD-209, and shall be located so that at least 1-inch clearance is maintained between slings and all exterior parts. Reinforcement may be fastened to members that shall withstand stresses, consistent with the amount and direction of pull. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.4.1.1. All slinging/tiedown provisions shall be labeled "LIFT", “TIEDOWN", or "LIFT TIEDOWN", as appropriate, in 1-inch (2.54 cm) high letters.

3.4.1.3 Rail transport. The pump assembly shall be rail transportable, in both a TRICON transport configuration and a stand-alone configuration, in the U.S. and North Atlantic Treaty Organization (NATO) countries without restriction. The pump assembly shall have a dimensional profile within the Gabarit International de Chargement (GIC) outline diagram (see MIL-STD-1366) when loaded aboard a 50-in high rail car.

3.4.1.3.1 Rail impact. The pump assembly, packed in TRICON for transport and stand-alone configuration, shall be capable of withstanding shock loads resulting from rail shipment or railroad car impacts as described in MIL-STD-810, without failure, damage, permanent deformation, or reduction in operational or design capability.

3.4.1.4 Highway transport. When transported on commercial or non-tactical military vehicles, the pump assembly shall be within the highway permit limits for all U.S. states and the general unrestricted transport in NATO countries IAW MIL-STD-1366 and AMOVP-1, not to exceed 4 meters in height (see 6.7).

3.4.1.5 Marine transport. The pump assembly shall be transportable by commercial ships and barges, and Army landing craft and barges.

3.4.1.6 Aircraft transport.

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3.4.1.6.1 Fixed-wing aircraft. The pump assembly shall be internally transportable by United States Air Force (USAF) C130 fixed-wing aircraft, and larger commercial and USAF aircraft (see MIL-STD-1791 for guidance).

3.4.1.6.2 Rotary-wing aircraft. The pump assembly shall be transportable by United States Army CH47 rotary-wing aircraft.

3.4.1.7 Forklift. The pump assembly shall include forklift pockets that are capable of safely lifting 10,000 lbs with 48-inch load centers and at least 72-inch tines, for use with standard Army 10,000 pound forklifts: All Terrain Lifter Army System (ATLAS) (TM 10-3930-673-10), and Articulated Frame Steer Forklift Truck (M10A) (TM 10-3930-643-10). The maximum spread, on center, of the forklift pockets shall be 28 inches, to interface with standard Army 4,000 pound forklifts.

3.4.2 NATO intervehicle receptacle. A receptacle conforming to NATO STANAG 4074 Type 1 (see also 6.8) with cover shall be provided. The receptacle shall be located in a protected position, and connected in parallel with the battery(s), to allow the pump assembly to accept electrical power from another nominal 24 VDC military system through a NATO intervehicle cable and plug assembly (see 6.8). The receptacle shall be labeled "24 VOLTS". The receptacle shall provide the capability of starting and running the pump assembly without pump assembly batteries.

3.4.3 Fuel. The pump assembly shall be capable of operating on all military and commercial kerosene based fuels conforming to, as a minimum, those listed below without restrictions or kits. JP-8 shall be the designated primary fuel for engine operation.

a. MIL-DTL-83133 (JP-8) (NATO F-34) (see 6.9)
b. MIL-DTL-5624 (JP-5) (NATO F-44) (see 6.9)
c. ASTM-D975 (Diesel-US commercial)
d. MIL-DTL-83133 (NATO F-35) (see 6.9)
e. ASTM-D1655 (Jet A-1)
f. AFLP-3747 (NATO F-24) (see 6.9)

3.4.4 Lubricants. The pump assembly, depending on the ambient temperature, shall be limited to the use of one or more of the following lubricating oils: MIL-PRF-2104, MIL-PRF-46167, and MIL-PRF-21260. Gear oil, if required, shall conform to SAE J2360, and grease to MIL-PRF-10924.

3.4.5 Dimensional and weight limits. Pump assembly shall weigh no greater than 4000 pounds and fit within the dimensions of 88.00 inches long (a retractable tow bar may be used), 72.00 inches wide, and 82.00 inches high. The pump assembly design shall permit loading into a TRICON IAW MIL-PRF-32349.

3.5 Ownership and support requirements.

3.5.1 Safety. All electrical wiring and any rotating or reciprocating parts shall be electrically and physically safe, and shall be guarded so as not to be a hazard to operating personnel and to minimize the hazard of fire in the event of a fuel spillage or leakage from hoses and connections. All electrical terminals shall be completely enclosed or insulated to prevent inadvertent contact by personnel or equipment that may cause arcing to occur. An emergency

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shutoff device shall be located at the operator station(s) for the assembly, to quickly and completely shut off/stop flow and the engine in case of accident, malfunction, or potential safety hazard. Danger or caution signs, labels and markings shall be used to warn of potential or specific hazards.

3.5.2 Human factors engineering. Human engineering criteria, principles, and practices shall be IAW MIL-STD-1472. The pump assembly shall be capable of being operated by the 5th percentile female to the 95th percentile male military personnel wearing gloves, arctic mittens, or equipped with Mission Orientated Protective Posture (MOPP) IV gear (see 6.6.2) without special tools or support equipment. Emphasis shall be given, but not limited to, general requirements, control and display integration, controls, labeling, maintainability, anthropometry, and hazard and safety criteria, as applicable.

3.5.3 Environmental hazard prevention. Provisions shall be made and included to prevent fluid leakage and safely contain leakage and spillage during all storage, transport and operational use of the pump assembly. The pump assembly shall comply with all US HAZMAT, OHSA, safety and transportation requirements, regarding prevention of inadvertent fuel discharge or leakage during operation, storage, and maintenance.

3.5.4 Lighting. All components of the pump assembly shall be effectively operable by the operator in reduced visibility, nighttime, and blackout conditions. All lighting shall be of the light emitting diode (LED) type in lieu of incandescent bulbs.

3.5.4.1 Lighting, non-tactical. Permanent or detachable lighting shall be provided to insure safe operation of the assembly in darkness and periods of reduced visibility.

3.5.4.2 Lighting, tactical/blackout operations. Blackout lighting provisions shall meet the following requirements:

a. A single blackout switch, labeled “BLACKOUT SWITCH”, shall control all light sources.
b. When blackout lighting is activated, ninety-five % of the light energy emitted by each light source shall be at wavelengths below 700 nanometers.
c. When blackout lighting is activated, all light sources shall be dimmed to 0.05 ft-Lamberts (fL), 0.17 candela per square meter, or less.

3.5.5 Grounding and bonding.

3.5.5.1 Bonding. The assembly shall provide complete electrical continuity throughout the unit, and shall include a metal-to-metal contact from the assembly to the ground. This includes all electrically conductive components of the unit that the operator and/or fuel may come in contact with. All bonding and/or grounding connections shall be mechanically secure and shall measure 1 (one) ohm or less.

3.5.5.2 Grounding rods, wires and clamps. For each pump assembly, a grounding rod shall be provided IAW drawing 13219E0462. A grounding wire with a minimum length of 20 feet shall be provided and equipped with a manual clamp requiring no tools for grounding the pump assembly during operation. The grounding wire shall be provided on a manual retracting reel mounted on the pump assembly. The pump assembly shall include stowage/storage provisions for all grounding components.

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3.5.6 Reliability. The mean-time-between-failures (MTBF) (see 6.6.3) of the assembly shall be not less than 200 hours at the 80% lower confidence level.

3.5.7 Maintainability. The assembly shall provide easy access to components requiring repair, replacement, routine or frequent maintenance or adjustment. The median time to repair (MedTTR) shall not exceed 1.5 hours for all unscheduled maintenance actions, and the maximum time to repair shall not exceed 3.0 hours for 90% of unscheduled maintenance actions (Max TTR90). All major assemblies and installed attachments requiring repair, replacement, and adjustment shall be accessible without the removal of other major components. Drain outlets shall be located for accessibility. All drain plugs and filler caps shall be durable and shall be permanently attached to the unit without impeding their removal. Corrosion resistance preventive maintenance shall be limited to routine washing, periodic inspection and repair of accident damage.

3.5.7.1 Maintenance ratio. The assembly shall have a maintenance ratio (see 6.6.4) of not greater than 0.07. Man-hours for repair of replaced components, and operator level, before-and-after-operation preventive maintenance checks and services (PMCS) are excluded. A maintenance schedule shall be established prior to start of any testing.

3.5.8 Operator equipment. All tools, special tools, and test equipment required to perform operator-level Preventive Maintenance Checks and Services (PMCS) (see 6.6.6) shall be provided with each system along with the required stowage space (see 3.5.9).

3.5.9 Storage container. A storage container shall be rigidly mounted on the assembly. The container shall be watertight when closed. The container shall be of a size capable of storing the items specified in 3.5.8 and 3.5.9.1.

3.5.9.1 Spares. Each pump assembly shall include the following accessories. Unless otherwise specified (see 6.2), each accessory shall be boxed or packaged separately and stored in the storage container of 3.5.9.

a. Set of engine oil filter(s).
b. Set of engine fuel filters(s).
c. Set of air filter(s).
d. Set of accessory/drive belt(s).
e. Eight (8) total quick-disconnect coupling gaskets IAW A-A-59326, of the appropriate sizes (see 3.3.2.3).
f. Funnel for priming port (see 3.3.2.1), with minimum 0.75 inch diameter spout.
g. US Army approved Technical Manual for the assembly.

3.5.10 Triple Container (TRICON). When specified (see 6.2), the pumping assembly shall include, and be packed in, a TRICON IAW MIL-PRF-32349.

3.6 Environmental requirements.

3.6.1 Operational temperature. The assembly shall operate as specified at any ambient temperature from -25°F to +120ºF.

3.6.2 Storage temperature. The assembly shall not be damaged by storage in any ambient temperature from -30ºF to +160ºF with humidity not to exceed 50%.

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3.6.3 Sand. The assembly shall perform as specified herein in a blowing sand environment when subjected to a minimum sand concentration of 0.062 ± 0.015 grams per cubic foot (g/ft^3) at a minimum wind velocity of 3540 feet per minute (ft/min).

3.6.4 Dust. The assembly shall perform as specified herein in a blowing dust environment when subjected to a minimum dust concentration of 0.3 ± 0.2 g/ft^3 at a minimum wind velocity of 1750 ft/min.

3.6.5 Humidity. The assembly shall perform as specified under relative humidity of up to 88%.

3.6.6 Solar radiation. The assembly shall perform as specified with up to 355 British thermal units per square foot per hour ($Btu/ft^2/hr$) of solar radiation.

3.6.7 Rain. The assembly shall perform as specified herein when subjected to a blowing rain environment of up to 4 in/hr. Pump assembly surfaces shall be designed to shed water and moisture regardless of incline.

3.6.8 Salt fog. The assembly shall withstand damage from being exposed to salt fog.

3.6.9 Altitude. The assembly, in storage configuration, shall withstand the low-pressure environment at a minimum altitude of 40,000 ft. The assembly shall perform as specified at altitudes up to and including 5,000 ft above sea level at a maximum temperature of 107°F.

3.6.10 Noise limits. Steady-state noise produced by the assembly shall not exceed 85 decibels (dB(A)) at the operator's position and at occasionally occupied positions (see 6.6.7). If noise levels are between 85 and 90 dB(A), or if procedures for noise suppression have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit has been obtained, noise hazard signs shall be prominently displayed on the equipment. Signs shall state:

DANGER

HEARING PROTECTION REQUIRED

WITHIN XX FEET

The word DANGER shall be on the first line and shall be white lettering on a red oval background which is inside a black rectangular panel in flat or lusterless colors. Other lettering shall be in black on a white background in lusterless colors. SAE AMS-STD-595 may be used for guidance. All lettering shall be readable at the maximum distance at which a noise level of 85 dB(A) is measured and this distance shall be inserted for XX.

3.6.11 Electromagnetic interference (EMI). The electromagnetic radiated interference and susceptibility characteristics of the assembly shall not exceed the limits specified in MIL-STD-461 for Army ground equipment or systems.

3.6.12 High-altitude Electromagnetic Pulse (HEMP). The assembly shall not exhibit any permanent malfunction or degradation of performance after being subjected to the default free-field electromagnetic pulse (EMP) environment IAW MIL-STD-464.

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3.6.13 Near Strike Lightning (NSL). The assembly shall not exhibit any permanent malfunction or degradation of performance after being subjected to the near lightning strike environment IAW MIL-STD-464.

3.6.14 Personnel Electrostatic Discharge (PESD). The assembly shall not exhibit any permanent malfunction or degradation of performance after being subjected to personnel electrostatic discharge IAW MIL-STD-464.

3.6.15 Helicopter Electrostatic Discharge (HESD). The assembly shall not exhibit any degradation of performance after being subjected to electrostatic discharge in the vertical lift mode of transport IAW MIL-STD-464.

3.6.16 Hazards of Electromagnetic Radiation to Fuel (HERF). Fuel(s) in the assembly shall not be inadvertently ignited by the electromagnetic environment (internal or external) IAW MIL-STD-464.

3.7 Personnel integration (MANPRINT) requirements. The assembly shall be operable and maintainable by the following military-trained personnel, as designated by DA Pam 611-21:

a. Military Occupational Specialty (MOS) 92F (Petroleum Supply Specialist).
b. MOS 92W (Water Treatment Specialist).
b. MOS 63J (Quartermaster and Chemical Equipment Repairman).
c. MOS 52D (Power Generator Repairman).

3.8 Identification, marking and information. Unless otherwise specified (see 6.2), identification, marking, information, and instruction plates shall be IAW MIL-STD-129 and MIL-STD-130.

3.8.1 Operating instructions and pump curves. One or two plates containing operating instructions and associated pump curves shall be mounted in a conspicuous location visible from the operator's position. The operating instructions shall be complete and brief, and shall describe procedures for pre-operation checks, starting, operating and stopping at ambient temperatures from - 25°F to +120°F.

3.8.2 Battery connection plate. A plate with a pictorial diagram showing the battery cable connections and instructions for removing and installing the battery shall be provided. The plate shall be mounted in, on, or at a conspicuous location near, the battery box (see 3.3.2.5.3).

3.8.3 Lubrication chart. A chart showing required lubrication, temperature ranges for lubrication, types of lubricants, quantities of lubricants and frequency of lubrication shall be mounted on the inside of the door to the control panel or instrument panel, or inside of the cover to the storage container (see 3.5.9). The chart shall also state types of coolant and fuel to be used.

3.8.4 Identification marking. The assembly shall be identified with plate(s) including at a minimum: specification number, design activity code, manufacturer and CAGE, contract number, date of manufacture, item nomenclature, serial number, NSN, weight, technical manual number, US military property identification, and pictorial line diagram(s) showing maximum dimensions and center-of-gravity location.

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3.9 Corrosion control performance. The assembly shall operate without corrosion affecting form, fit, or function during a 20-year service life. Service life can include varying or extended periods in corrosive environments involving one or more of the following: high humidity, salt spray, road de-icing agents, gravel impingement, atmospheric contamination, and temperature extremes. During the 20-year service life, surface corrosion, which may include red, black, and/or white corrosion products, shall be a maximum of 1% of the surface. The loss of the original thickness of the base metal shall be no greater than 2% or 0.001 inch, whichever is less.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), all external surfaces of the pump assembly except as noted below, regardless of the material selected, shall have a finish coat of Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant (Waterborne CARC) paint IAW MIL-DTL-64159. The color shall be Desert Tan 686A, color number 33446 IAW SAE AMS-STD-595.

a. The color of all other surfaces, to include those within a housing, those behind insulation material, and the insulation retainer, if used, shall be Desert Tan 686A, color number 33446 or Black, color number 37030 of SAE AMS-STD-595 or the manufacturer standard color, if approved by the Government.
b. The following items shall not be painted: terminal wiring connections, governor linkage, instruction diagrams and plates, rectifiers, relays, switches, circuit interrupters, instrumentation, rubber, lubrication fittings, hoses, nozzles, dry break couplings, and all other parts whose operation or function would be adversely affected by paint. Insulation material shall be painted unless the sound absorbing characteristics of the material are compromised.
c. Identification plates shall be installed following application of the final finish coat.
d. CARC paint shall not be applied to surfaces that may exceed temperatures of 400°F.

3.11 Welding. All welding shall be in accordance with AWS D1.1/D1.1M, AWS D1.2/D1.2M, AWS D1.3/D1.3M and the ASME Boiler and Pressure Vessel Code.

3.12 NBC decontamination survivability. The pump assembly shall be hardened against the effects of NBC contamination and the decontamination process (see 3.10). The pump assembly shall be NBC contamination survivable and capable of operating in an NBC environment in accordance with FM 3-5. The design of the system components shall minimize the collection and retention of contaminants and decontaminates. The system and components shall be decontaminable to negligible risk levels as defined in Army-approved NBC contamination survivability criteria and AR 70-75. All seals shall be resistant to the NBC agents and compatible with decontamination procedures and material.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article test (FAT) (see 4.1.1).
b. Conformance inspection (CI) (see 4.1.2).

4.1.1 First article test (FAT). Unless otherwise specified (see 6.2), the contractor shall submit a first article sample consisting of three complete assemblies packed in their TRICONs.

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FAT shall be performed by the contractor on the first production samples. Three assemblies are suggested for FAT sample to allow for parallel and concurrent testing, in order to expedite test schedule and reduce costs. FAT shall include the test conditions, examinations, tests and certifications in 4.1 through 4.12. Modeling and simulation is prohibited for all tests, unless otherwise specified herein. Nonconformance to any requirement herein or contractual requirement, or failure of any required test or examination shall be cause for rejection by the Government (see 3.1, 6.2, and 6.3).

4.1.1.1 Examination. Prior to start of FAT, each pump assembly submitted for FAT sample shall be examined as specified in 4.1.2.

4.1.1.2 Certificates of Conformance (CoC). CoCs required as part of the FAT shall be included in the FAT final report, and IAW contractor format with sufficient supporting technical information to insure adequate evaluation and comment by the Government. CoCs may be required for selected requirements either prior to or during Government conducted testing for safety, HAZMAT, and operational requirements in order to assure safe operation by test site personnel.

4.1.2 Conformance inspection (CI).

4.1.2.1 Examination. Each pump assembly production unit shall be examined as specified in Table I. Any nonconformance or missing requirement revealed by the examination shall be cause for rejection.

Table I. Examination Schedule.

Number	Exam Description	Method of Inspection
101	Materials not as specified (see 3.2)	Visual/Doc 1/
102	Operating requirements not as specified (see 3.3)	Visual/Doc
103	Threaded parts not as specified (see 3.3.2)	SIE 2/Doc
104	Pump housing not as specified (see 3.3.2.1)	Visual
105	Piping, fittings and valves not as specified (see 3.3.2.2)	Visual/Doc
106	Suction and discharge components not as specified (see 3.3.2.3)	Visual/Doc
107	Engine not as specified (see 3.3.2.4)	Visual/Doc
108	Air cleaner not as specified (see 3.3.2.4)	Visual/Doc
109	Fuel filters not as specified (see 3.3.2.4)	Visual/Doc
110	Exhaust system not as specified (see 3.3.2.4)	Visual/Doc
111	Engine oil drain tube not as specified (see 3.3.2.4)	Visual/Doc
112	Engine oil sample valve not as specified (see 3.3.2.4)	Visual/Doc
113	Cooling system not as specified (see 3.3.2.4)	Visual/Doc
114	Engine fan belt guard not as specified (see 3.3.2.4)	Visual/Doc
115	Oil dipstick not as specified (see 3.3.2.4)	Visual
116	Electrical system and components not as specified (see 3.3.2.5)	Visual/Doc
117	Batteries not as specified (see 3.3.2.5.1)	Visual/Doc
118	Battery cables not as specified (see 3.3.2.5.2)	Visual

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Table I. Examination Schedule – Continued.

119	Battery box not as specified (see 3.3.2.5.3)	Visual/Doc
120	Control and instrument panel(s) not as specified (see 3.3.2.6)	Visual/Doc
121	Control panel not as specified (see 3.3.2.6.1)	Visual/Doc
122	Control panel components not as specified (see 3.3.2.6.1.1)	Visual/Doc
123	Engine/pump protection not as specified (see 3.3.2.6.1.2)	Visual/Doc
124	Shutdown annunciation not as specified (see 3.3.2.6.1.3)	Visual/Doc
125	Instrument panel not as specified (see 3.3.2.6.2)	Visual/Doc
126	Pressure lines not as specified (see 3.3.2.6.3)	Visual
127	Automatic pump controller not as specified (see 3.3.2.7)	Visual/Doc
128	Manual mode not as specified (see 3.3.2.7.1)	Visual/Doc
129	Automatic speed control mode not as specified (see 3.3.2.7.2)	Visual/Doc
130	Automatic pressure control mode not as specified (see 3.3.2.7.3)	Visual/Doc
131	Suction pressure limit override not as specified (see 3.3.2.7.4)	Visual/Doc
132	Battle Short bypass not as specified (see 3.3.2.7.5)	Visual/Doc
133	Fuel system not as specified (see 3.3.2.8)	Visual
134	Fuel reservoir not as specified (see 3.3.2.8.1)	Visual/Doc
135	Refueling not as specified (see 3.3.2.8.2, Type I only)	Visual
136	Chassis not as specified (see 3.3.2.9)	Visual/Doc
137	Tires and wheels not as specified (see 3.3.2.9.1)	Visual
138	Brakes not as specified (see 3.3.2.9.2)	Visual/Doc
139	Towing extension not as specified (see 3.3.2.9.3)	Visual/Doc
140	Stabilizing legs not as specified (see 3.3.2.9.4)	Visual
141	Lights, reflectors or connector not as specified (see 3.3.2.9.5)	Visual/Doc
142	Transportability not as specified (see 3.4.1)	Doc
143	Tiedown provisions not as specified (see 3.4.1.1)	Visual/Doc
144	Slinging provisions not as specified (see 3.4.1.2)	Visual/Doc
145	NATO receptacle not as specified (see 3.4.2)	Visual/Doc
146	Forklift provisions not as specified (see 3.4.1.7)	SIE/Doc
147	Lubricants not as specified/compromised (see 3.4.4)	Visual/Doc
148	Dimensional and weight limits not as specified (see 3.4.5)	SIE
149	Safety characteristics not as specified (see 3.5.1)	Visual/Doc
150	Human factors engineering not as specified (see 3.5.2)	Visual/Doc
151	Environmental hazard prevention not as specified (see 3.5.3)	Visual/Doc
152	Lighting and controls not as specified (3.5.4)	Visual/Doc
153	Grounding equipment not as specified (see 3.5.5.2)	Visual/Doc
154	Reliability not as specified/compromised (see 3.5.6)	Visual/Doc
155	Maintainability not as specified/compromised (see 3.5.7)	Visual/Doc
156	Operator equipment missing (see 3.5.8)	Visual/Doc
157	Storage and/or spares missing or not as specified (see 3.5.9)	Visual
158	TRICON missing or not as specified (see 3.5.10)	Visual
159	Environmental requirements not as specified (see 3.6)	Visual/Doc
160	MANPRINT requirements not as specified (see 3.7)	Visual/Doc
161	Identification/marking/information not as specified (see 3.8)	Visual/Doc

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Table I. Examination Schedule – Continued.

162	Corrosion control not as specified/compromised (see 3.9)	Visual/Doc
163	Treatment and painting not as specified/compromised (see 3.10)	Visual/Doc
164	Welding not as specified (see 3.11)	Visual/Doc
165	NBC survivability requirements not as specified (see 3.12)	Visual/Doc

1/ Doc = Documentation
2/ SIE = Standard Inspection Equipment

4.2 Test conditions and schedule.

4.2.1 Test conditions. Unless otherwise specified herein, all testing and examinations shall be conducted at ambient atmospheric pressure and humidity, preferably sheltered from sunlight, rain/snow, and wind. All performance data shall be corrected to standard sea level conditions of 29.92 inch of mercury (barometric pressure) and water at 60°F. JP-8 shall be the designated primary fuel for the engine.

4.2.2 Schedule and sequence of tests. Schedule and sequence of contractor-conducted tests and examinations shall be made by the contractor, and provisions shall be made to insure that each item of the schedule and sequence shall not have a detrimental effect or preclude the performance of a subsequent test/examination.

4.2.3 Water substitution. Water is the preferred test fluid to be pumped, but JP-8 may be used for any test. Values shall be corrected for each of the required fuels in 3.4.3.

4.3 Design and operating requirements verification. The requirements of 3.2 and 3.3 shall be demonstrated through one or more of the following: successful completion of the required FAT of the pump assembly, continuing compliance with assembly CI requirements, the absence of non-compliance and material faults during the performance of the contract, or the (projected) completion of the service life requirement(s). Failure to comply with all the requirements of 3.2 and 3.3 and any of their subparagraphs or specifications shall constitute failure of these tests.

4.3.1 Pump assembly performance verification.

4.3.1.1 Hydrostatic pressure. To verify conformance to 3.3.1.1, a non-operating complete pump assembly shall be subjected to a hydrostatic pressure of at least 225 psig, for a period of not less than 5 minutes. The test shall be performed with the valves open and with couplers' dust covers installed. Gauges may be excluded from testing. Rupture or distortion to any component, and any leakage through the assembly, fittings, piping, or joints shall constitute failure of this test.

4.3.1.2 Capacity. To verify conformance to 3.3.1.2 and 3.3.1.2.1, a pump assembly shall be operated under the conditions as specified in 4.2 and tilted front to back and side to side at any angle of 15-20 degrees from the horizontal. Data relative to head and capacity of the pump assembly, and speed (rpm) of the engine shall be measured IAW ANSI/HI 14.6. Delivery of less than the capacities specified in 3.3.1.2 or failure of the engine to meet any of the operational requirements herein shall constitute failure of this test.

4.3.1.3 Pump priming. To verify conformance to 3.3.1.3, a pump case shall be filled with water at conditions as specified in 4.2 and a priming test conducted with a static suction lift

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equivalent to at least 10 ft of water using six-inch diameter suction hose. Inability of the assembly to prime and deliver full capacity in 2 minutes shall constitute failure of this test.

4.3.1.4 Slope capability. To verify conformance to 3.3.1.4, the pump assembly, in its transport mode, shall be towed upgrade, then downgrade, on a ramp of not less than a 30% slope, in both forward and reverse directions, by a designated prime mover (see 3.3.1.5.1). During each negotiation the pump assembly shall be observed for contact with the ground surface, other than tires. The pump assembly shall then be towed, transversely, along a side slope of not less than 20%, at a speed not greater than 8 mph. The procedure shall be repeated in the opposite direction. Any contact of the assembly with the ground other than the tires, or contact of the assembly with the tires, or the inability to traverse any slope shall constitute failure of this test.

4.3.1.5 Mobility. Mobility testing shall be conducted for the minimum distances and speeds as specified in 4.3.1.5.2 or 4.3.1.5.3, as appropriate, and shall be conducted at a site approved by the Government. After the conclusion of all test conditions in 4.3.1.5.2, the pump assembly shall be operated at rated capacity for not less than one hour. Any malfunction, permanent deformation, damage, or failure of the pump assembly to operate as specified in 3.3.1.2 shall constitute failure of this test.

4.3.1.5.1 Prime movers. To verify conformance to 3.3.1.5.1, CoCs with supporting engineering documentation attached shall be provided. Simulation is acceptable and desirable. Absence of any CoC or supporting documentation shall constitute failure.

4.3.1.5.2 Mobility. The pump assembly shall be towed over the following conditions:

a. Improved (paved) roads at an average speed of 20 ± 3 miles per hour (mph), maximum speed of 25 mph, for a minimum of 100 miles.

b. Unimproved (unpaved) roads at an average speed of 10 ±2 mph, maximum speed of 12 mph, for a minimum of 100 miles.

c. Over cross country terrain at an average speed of 8 ±1 mph, maximum speed of 10 mph, for a minimum of 50 miles.

4.3.1.5.3 Mobility in transport configuration. The pump assembly shall be packed in its TRICON and transported over the following conditions:

a. Improved (paved) roads at an average speed of 50 ± 5 miles per hour (mph), maximum speed of 55 mph, for a minimum of 200 miles.

b. Unimproved (unpaved) roads at an average speed of 20 ± 3 mph, maximum speed of 25 mph, for a minimum of 500 miles.

c. Over cross country terrain at an average speed of 8 ±1 mph, maximum speed of 10 mph, for a minimum of 300 miles.

4.3.1.6 Drop. To verify conformance to 3.3.1.6, the pump assembly, balanced on only its own two wheels, shall be allowed to fall forward onto the lunette, and backward until it contacts the ground. The height of the forward drop shall be 26 inches, and shall be measured from the bottom of the lunette to the ground. The ground surface shall be hard-packed gravel or hard-packed dirt. A total of four forward drops shall be made, and one backward drop/contact. All drops shall be conducted in the towing configuration, with jackstands fully retracted and sandshoes removed, if applicable. If a retractable tow bar is present, the drops shall be conducted with the tow bar extended in towing configuration. The pump assembly shall then be

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inspected for damage or broken components and operated for not less than one-half hour. Any breakage, permanent deformation or failure of the assembly to operate as specified in 3.3.1.2 shall constitute failure of this test. Nicks, dents, scrapes, or a marred finish shall not be considered defects.

4.3.1.7 Fording capability. To verify conformance to 3.3.1.7, the pump assembly, in its towing mode, shall be towed through water at depth of at least 30 in and at a speed not greater than 8 mph. The pump assembly shall be operated within 5 minutes of extraction from the water and inspected for damage. Failure of the pump assembly to operate as specified in 3.3.1.2 shall constitute failure of this test.

4.3.2 Pump assembly characteristics and components.

4.3.2.1 Pump housing. To verify conformance to 3.3.2.1, visually verify pump housing condition, mandatory interface requirements, components and suitability for all intended uses. Additionally, a pump shall be filled with water, and the drain plug removed. Record the amount of time for the pump to fully drain. Failure to provide a pump housing IAW the requirements of 3.3.2.1, or the inability of the pump to drain in five (5) minutes or less shall constitute failure of this test.

4.3.2.2 Piping and fittings. To verify conformance to 3.3.2.2, CoCs shall be provided for all piping, fittings and vents. Verify count, condition, location and mandatory interface requirements, and suitability for all intended uses. Absence of any CoC, missing components or failure to meet any requirement of 3.3.2.2 shall constitute failure.

4.3.2.3 Suction and discharge components. To verify conformance to 3.3.2.3, CoCs shall be provided. Verify count, condition, mandatory interface requirements, and suitability for all intended uses. Absence of any CoC or failure to comply with the requirements of 3.3.2.3 shall constitute failure.

4.3.2.4 Engine. To verify conformance to 3.3.2.4, CoCs shall be provided, and the pump assembly shall be examined for the presence of the minimum equipment specified in 3.3.2.4. Verify condition and suitability for all intended uses. Absence of any CoC or failure to comply with the requirements of 3.3.2.4 shall constitute failure.

4.3.2.5 Electrical system. To verify conformance to 3.3.2.5 and all subparagraphs, visual confirmation and CoCs shall be provided for the batteries, battery cables and box. Verify condition, mandatory interface requirements, and suitability for all intended uses. Absence of any CoC or failure to comply with any requirement of 3.3.2.5 or any subparagraph shall constitute failure.

4.3.2.6 Control and Instrument panels. To verify conformance to 3.3.2.6, CoCs shall be provided, and the pump assembly control and instrument panels shall be examined for the presence of the minimum equipment and features specified in 3.3.2.6 and all subparagraphs. Failure of the control panel to be replaced to a fully operational condition, without the use of special tools, within the allowed time, shall be considered a failure. Absence of any CoC or failure to comply with the requirements of 3.3.2.6 and all subparagraphs shall constitute failure.

4.3.2.7 Automatic pump controller. To verify conformance to 3.3.2.7 and all subparagraphs, visual confirmation and CoCs shall be provided for the automatic pump

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controller, modes of operation, override, and bypass. Verify condition and suitability for all intended uses. Absence of any CoC or failure to comply with any requirement of 3.3.2.7 or any subparagraph shall constitute failure.

4.3.2.8 Fuel system. To verify conformance to 3.3.2.8, CoCs shall be provided, and the pump assembly fuel system shall be examined for the presence of the features and markings specified in 3.3.2.8. Absence of any CoC or failure to comply with the requirements of 3.3.2.8 shall constitute failure.

4.3.2.9 Chassis. To verify conformance to 3.3.2.9 and all subparagraphs, CoCs shall be provided, and the pump assembly shall be examined for the presence of the equipment specified in 3.3.2.9 and all subparagraphs. Absence of any CoC or failure to comply with any requirement of 3.3.2.9 or any subparagraph shall constitute failure.

4.4 Interface requirements verification.

4.4.1 Transportability. The pump assembly shall be verified, through demonstrations and tests, the requirements of 3.4.1. In addition to the actual testing required by this paragraph, a CoC shall be provided to certify compliance and system compatibility with all the transport configurations listed in 3.4.1. Absence of the CoC, failure of the CoC to certify all the requirements of 3.4.1, or failure of any test or demonstration shall constitute failure.

4.4.1.1 Tiedown provisions. To verify conformance to 3.4.1.1, CoC shall be provided to verify the design, material, test and process requirements of MIL-STD-209. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.1 shall constitute failure.

4.4.1.2 Slinging provisions. To verify conformance to 3.4.1.2, CoC shall be provided to verify the design, material, test, and process requirements of MIL-STD-209. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.2 shall constitute failure.

4.4.1.3 Rail transport. To verify conformance to 3.4.1.3, the pump assembly, in its TRICON shipment configuration, and stand-alone configuration, shall be examined and analyzed to determine whether the system can be safely transported by rail in North America, Europe, and Korea without restrictions and has a dimensional profile within the five rail clearance diagrams required when loaded on a 50-inch (127 cm) high rail car. CoC shall be provided to verify all of the design requirements of 3.4.1.3. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.3 shall constitute failure.

4.4.1.3.1 Rail impact test. To verify conformance to 3.4.1.3.1, the pump assembly, mounted on a rail car in its TRICON shipment configuration, and in its stand-alone configuration, shall be tested IAW MIL-STD-810, Procedure VII of Method 516 (rail impact). Any pump assembly component that breaks free, loosens or shows any sign of permanent deformation or reduction of performance as a result of the impact test series shall constitute failure.

4.4.1.4 Highway transport. To verify conformance to 3.4.1.4, the pump assembly shall be verified for compliance through detailed design analysis, examinations, and conformance certification to the requirements of 3.4.1.4. Examination and analysis shall be carried out to verify the pump assembly can meet the US and NATO countries highway transport limits for all

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configurations as specified in 3.4.1.4. Inability of the pump assembly to meet the requirements as specified shall constitute failure.

4.4.1.5 Marine transport. To verify conformance to 3.4.1.5, the pump assembly shall be certified as safely transportable by all the watercraft specified in 3.4.1.5 through design analysis, examinations, modeling, or simulation, and a CoC shall be provided as part of the final test report. Inability to meet the requirements as specified for all configurations in 3.4.1.5 shall constitute failure.

4.4.1.6 Aircraft transport. The pump assembly shall be certified by CoC provided prior to the start of FAT as safely transportable by all the aircraft types specified in 3.4.1.6.1 and 3.4.1.6.2 through the use of MIL-STD-1366 and recommendations in MIL-STD-1791, as applicable; design analysis; examinations and test; modeling and simulation; and observations.

4.4.1.6.1 Fixed-wing aircraft. To verify conformance to 3.4.1.6.1, examination and analysis shall be conducted to verify the pump assembly can be internally transportable by USAF C130 and larger commercial fixed wing aircraft. A CoC shall be provided as part of the final test report for verification of the requirements of 3.4.1.6.1. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.6.1 shall constitute failure.

4.4.1.6.2 Rotary-wing aircraft. To verify conformance to 3.4.1.6.2, examination and analysis shall be carried out to verify the pump assembly can be safely transported, by US Army CH47 rotary-wing aircraft. A CoC shall be provided as part of the final test report for verification of the requirements of 3.4.1.6.2. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.6.2 shall constitute failure.

4.4.1.7 Forklift. To verify conformance to 3.4.1.7, examination and analysis shall be carried out to verify the pump assembly can be safely handled by the specified forklifts. A CoC shall be provided as part of the final test report for verification of the requirements of 3.4.1.7. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.1.7 shall constitute failure.

4.4.2 NATO intervehicle receptacle. To verify conformance to 3.4.2, the pump assembly shall be examined for the presence, proper location, labeling, and installation of the receptacle and its attaching connections/wiring. A CoC shall be provided as part of the final test report for verification of the requirements of 3.4.2. Absence of the CoC or failure of the CoC to certify all the requirements of 3.4.2 and STANAG 4074 shall constitute failure.

4.4.3 Fuel. To verify conformance to 3.4.3, the designated primary fuel (JP-8) shall be utilized by the engine during actual pumping and engine operation during RAM and other performance tests, except as specified in this paragraph and 4.2.3. A CoC shall be provided as part of the final test report verifying the suitability, without restrictions or kits, of the remaining engine fuels required, but not actually demonstrated, during the RAM testing. Absence of the CoC, failure of the CoC to certify all the requirements of 3.4.3, or failure of any test shall constitute failure.

4.4.4 Lubricants. To verify conformance to 3.4.4, lubricants listed in 3.4.4 shall be used during both FAT and CI based on applicable ambient temperatures. Design analysis and CoC by each engine and pump manufacturer for acceptability of use shall be provided for all lubricants

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listed in 3.4.4 as part of the final test report. Failure to use the required lubricants, inability of the components to perform without restrictions with the required lubricants, or absence of any required CoC shall constitute failure.

4.4.5 Dimensional and weight limits. The external dimensions of the pump assembly, in storage configuration, shall be measured using standard inspection equipment. The pump assembly, in operational configuration but without fuel, shall be placed on a scale capable of weighing in 5 lb increments and the weight determined to the nearest 5 lb. Any dimension or weight greater than that specified in 3.4.5 shall constitute failure of this test.

4.5 Ownership and support requirements verification.

4.5.1 Safety. To verify conformance to 3.5.1, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Failure to provide the CoC in the FAT final report or comply with the requirements of 3.5.1 during FAT or CI shall constitute failure.

4.5.2 Human factors engineering. To verify conformance to 3.5.2, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Failure to provide the CoC in the FAT final report or comply with the requirements of 3.5.2 during FAT or CI shall constitute failure.

4.5.3 Environmental hazard prevention. To verify conformance to 3.5.3, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Failure to provide the CoC in the FAT final report or comply with the requirements of 3.5.3 during FAT or CI shall constitute failure.

4.5.4 Lighting. To verify conformance to 3.5.4 and all subparagraphs, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Absence of specified tactical or non-tactical lighting devices, failure to provide the CoC in the FAT final report or failure to comply with any requirement of 3.5.4 or any subparagraph during FAT or CI shall constitute failure.

4.5.5 Grounding and bonding. To verify conformance to 3.5.5 and all subparagraphs, the pump assembly shall be examined for the presence, and secure and safe stowage/storage of the grounding components. A CoC shall be provided as part of the final test report for verification of the requirements of 3.5.5 and all subparagraphs. Absence of the CoC or failure of the CoC to certify all the requirements of 3.5.5 and all subparagraphs shall constitute failure.

4.5.6 Reliability. To verify conformance to 3.5.6, a minimum of three assemblies shall be operated at various loads up to and including rated capacity (see 3.3.1.2) under the standard conditions specified in 4.2. Each pump assembly shall be continuously operated for not less than 9 hours out of every 24 hours. One assembly shall be operated as a stand-alone in manual mode, drawing directly from the pumpage source and discharging back to storage. The other two assemblies shall be operated in series, with the first pump in series operating in manual mode and drawing from the pumpage source, and the following assembly operating in automatic mode. The pump operating in automatic mode shall be randomly operated in each of the automatic modes. The order of the assemblies shall be rotated approximately every 100 hours of operation, such that each assembly shall be subjected to each location in the configuration, for example

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stand-alone, first in series, and second in series. Each pump assembly shall be operated for a minimum of 1.5 times the specified MTBF. Failure to demonstrate the MTBF requirement in 3.5.6 to the 80% lower confidence level shall constitute failure of this test.

4.5.7 Maintainability. To verify conformance to 3.5.7, for each pump assembly, total active maintenance man-hours (broken out by scheduled and unscheduled maintenance) and total unscheduled maintenance actions shall be recorded during the reliability test specified in 4.5.6. Calculating from this data, a median time to repair (for all unscheduled maintenance actions) greater than 1.5 hours, or maximum time to repair (for 90% of unscheduled maintenance actions) greater than 3 hours or nonconformance to any requirement of 3.5.7 shall constitute failure.

4.5.7.1 Maintenance ratio. To verify conformance to 3.5.7.1, the maintenance ratio shall be computed during FAT, and a maintenance ratio of greater than 0.07 or failure to provide a maintenance schedule prior to the start of any testing shall constitute failure.

4.5.8 Operator equipment. To verify conformance to 3.5.8, the pump assembly shall be continuously evaluated during the conduct of the FAT for compliance with the requirements of 3.5.8, and this evaluation, including any deficiencies and corrective actions, shall be included in the FAT final report. Failure to comply with any requirement of 3.5.8, missing or damaged support items, or degradation of pump assembly performance capability due to inability to conduct PMCS or maintain or repair at the prescribed levels, or insufficient tools or types of tools to complete required maintenance activities or repairs, shall constitute failure.

4.5.9 Storage container. To verify conformance to 3.5.9 and all subparagraphs, CoCs shall be provided, and the pump assembly shall be examined for the presence of the equipment specified in 3.5.9 and all subparagraphs. Absence of any CoC or failure to comply with any requirement of 3.5.9 or any subparagraph shall constitute failure.

4.5.10 Triple Container (TRICON). To verify conformance to 3.5.10, the pump assembly shall be examined for the presence of the equipment specified in 3.5.10. Failure to comply with any requirement of 3.5.10 shall constitute failure.

4.6 Environmental requirements verification. To verify conformance to 3.6, the pump assembly shall be subjected to the following environmental tests in its operational mode, except when otherwise specified. Unless otherwise specified herein, all test procedures shall be IAW MIL-STD-810. On a continuing basis during and at the conclusion of each procedure of this test, the pump assembly shall be examined for any damage, leakage, failure to operate as required, or degradation of operational capabilities. Any damage, leakage, failures, inability of the pump assembly to operate, reduction in assembly operational capabilities, or failure to perform to the requirements of each test shall constitute failure of these tests.

a. At the conclusion of each test and prior to the start of the next test method, the (engine) air filters/filtered vents shall be removed, examined, and the results recorded for adequacy of filtration, proper sealing, and any evidence of failure or significant degradation. The air filters/filtered vents may then be replaced with new filter elements prior to the start of the next test, or cleaned, if applicable.

4.6.1 High operational and storage temperatures. To verify conformance to 3.6.1 and 3.6.2, the pump assembly shall be consecutively tested IAW Procedure I (Storage) and Procedure

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II (Operation) of Method 501. Temperature sensors shall be located in all locations and in operational fluids that may be harmed by exposure to high temperatures. As a minimum, pumpage temperature, engine oil temperatures, engine exhaust temperatures, pump casing temperatures, and housing temperatures shall be monitored and recorded at regular intervals. Test temperatures shall be as specified in 3.6.1 and 3.6.2. Test duration shall be seven (7) 24-hour cycles for Procedure I and constant temperature exposure for Procedure II. The pump assembly shall be operated at rated flow until all recorded temperatures have stabilized or a minimum of 4 hours. Failure of the pump assembly to operate as specified, or any damage or degradation of performance as a result of exposure to the temperature extremes, shall result in failure of this test.

4.6.2 Low operational and storage temperatures. To verify conformance to 3.6.1 and 3.6.2, the pump assembly shall be tested IAW Procedures I, II, and III of Method 502. Temperature sensors shall be located in all locations and in operational fluids that may be harmed by exposure to low temperatures. As a minimum, pumpage temperature, engine oil temperatures, engine exhaust temperature, pump casing temperature, and housing temperatures shall be monitored and recorded at regular intervals. Test temperatures shall be as given in 3.6.1 and 3.6.2. Test duration is not less than 4 hours for storage, and not less than 4 hours for operation. When water is used as the test fluid, or pumpage, then measures shall be taken during operation to prevent the water from dropping below the freezing point. Failure of the pump assembly to operate as specified, or any damage or degradation of performance as a result of exposure to the temperature extremes, shall result in failure of this test.

4.6.3 Sand. To verify conformance to 3.6.3, the pump assembly shall be tested IAW Procedure II of Method 510. The control panel door shall by fully open for the duration of this test. The pump assembly shall be subjected to blowing sand condition for a minimum total of 360 minutes, 90 minutes per side. The blowing sand concentration shall be maintained at 0.0623 ± 0.015 g/ft^3. The air velocity shall be 3540 ft/min for blowing sand. The system shall be operated during the last hour of the test. Upon test completion the pump assembly shall be started and operated for a period of not less than 30 minutes. Failure of the pump assembly to operate as specified shall result in failure of this test.

4.6.4 Dust. To verify conformance to 3.6.4, the pump assembly shall be tested IAW Procedure I (Dust) of Method 510. The control panel door shall by fully open for the duration of this test. The pump assembly, in full operational mode (running), shall be subjected to blowing dust condition for a minimum of 6 hours. The dust concentration for the blowing dust test shall be maintained at 0.3 ± 0.2 g/ft^3. The air velocity shall be 1750 ft/min for blowing dust. At the conclusion of this test, the pump assembly shall be inspected for visible signs of damage and operated for not less than 30 minutes. Failure of the pump assembly to operate as specified shall result in failure of this test.

4.6.5 Humidity. To verify conformance to 3.6.5, the pump assembly shall be tested IAW Method 507. The control panel door shall by fully open for the duration of this test. The test shall be conducted for 5 cycles IAW figure 507-1, at a maximum relative humidity of 88%. Failure of the pump assembly to operate as specified shall result in failure of this test.

4.6.6 Solar radiation. To verify conformance to 3.6.6, the pump assembly shall be tested IAW Procedure I of Method 505. The control panel door shall by fully open for the duration of

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this test. Test duration shall be not less than 5 cycles of 24 hours each. Temperature sensors shall be located in critical areas of highest temperature. The diurnal cycle shall be for hot-dry conditions. Upon completion of the test the pump assembly shall be examined for damage and operated for not less than 30 minutes. Failure of the pump assembly to operate as specified shall result in failure of this test.

4.6.7 Rain. To verify conformance to 3.6.7, the pump assembly shall be tested IAW Procedure I (Rain and Blowing Rain) of Method 506. The control panel door shall by fully open for the duration of this test. The water distribution device shall deliver water at a pressure from 30 to 40 psi and flow rate of 5 gallons per hour (gph) for a 30-minute period per side. A rainfall rate of up to 4 inch per hour impinging on the system at angles from the vertical up to 45° shall be used. Test item temperature shall be a minimum of 20°F higher than the rain temperature at the beginning of each exposure period. A wind velocity of 40 mph shall be used for this test. On completion of exposure, the pump assembly shall be operated at full capacity for not less than 30 minutes. Failure of the pump assembly to operate as specified shall result in failure of this test.

4.6.8 Salt fog. To verify conformance to 3.6.8, the pump assembly shall be tested IAW Method 509. The control panel door shall by fully open for the duration of this test. Duration of this test shall be 48 hours with the pump assembly in its operating configuration. Upon completion of the test the pump assembly shall be examined for damage and operated for not less than 30 minutes. Failure of the pump assembly to operate as specified or any visible surface corrosion of any metal parts, coated or uncoated, shall result in failure of this test.

4.6.9 Altitude. To verify conformance to 3.6.9, the pump assembly shall be tested IAW Procedure I (Storage/Air Transport) and Procedure II (Operation/Air Carriage) of Method 500. Procedure I shall be conducted for a 1-hour storage period at an altitude of not less than 40,000 ft above sea level. Procedure II shall be conducted for a 1-hour storage period at an altitude of not less than 5,000 ft above sea level with the pump assembly operating at rated capacity for not less than 1 hour. Failure of the pump assembly to operate as specified, or any damage as a result of exposure to the altitude extremes, shall result in failure of this test. Degradation of performance greater than ten percent (10%) below the rated performance of 3.3.1.2 (flow and/or pressure) shall be considered a failure.

4.6.10 Noise limits. To verify conformance to 3.6.10, noise levels shall be measured when the pump assembly is operating at rated capacity. Noise levels shall be recorded at the operator's position and occasionally occupied positions (see 6.6.7). Distance from the noise source shall also be recorded at not fewer than 12 equal (horizontal) arc increments at which readings of 85 dB(A) are measured. One measurement shall be the greatest distance from the noise source at which 85 dB(A) is measured. Noise levels shall be provided as dB(A) level. NOTE: Standard test conditions shall apply (see 4.2). If applicable, the presence of required noise hazard signs shall be verified. Failure to comply with any requirement 3.6.10 shall constitute failure of this test.

4.6.11 EMI. To verify conformance to 3.6.11, the pump assembly shall be tested IAW MIL-STD-461, requirements RE102 and RS103 for Army ground equipment. Failure to comply with MIL-STD-461 shall constitute failure of this test.

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4.6.12 HEMP. To verify conformance to 3.6.12, the pump assembly shall be tested IAW MIL-STD-461, requirement RS105. Failure to comply with MIL-STD-464, any permanent damage to the assembly, or failure to restart the pump assembly within fifteen (15) minutes if it stops operating, shall constitute failure of this test.

4.6.13 NSL. To verify conformance to 3.6.13, the pump assembly in its operational configuration shall tested IAW MIL-STD-464 to a 200,000 Amp strike at 100 meters (m), with an electrical field strength of 120,000 volts per meter (V/m). Failure to comply with MIL-STD-464, any permanent damage to the assembly, or failure to restart the pump assembly within fifteen (15) minutes if it stops operating, shall constitute failure of this test.

4.6.14 PESD. To verify conformance to 3.6.14, the pump assembly in its operational configuration shall be tested IAW MIL-STD-464 to discharges at multiple locations accessible to the operator. Discharges shall be 25,000 ± 500V, capacitance of 500 ± 5% picofarads, and series resistance of 500 ± 5% ohms. Failure to comply with MIL-STD-464, any permanent damage to the assembly, or failure to restart the pump assembly within fifteen (15) minutes if it stops operating, shall constitute failure of this test.

4.6.15 HESD. To verify conformance to 3.6.15, the pump assembly in its vertical lift configuration shall be tested IAW MIL-STD-464 to discharge(s) of 300 kilovolts, capacitance of 1000 ± 5% picofarads, and maximum resistance of one ohm. Failure to comply with MIL-STD-464, any permanent damage to the assembly, or failure to start the pump assembly, shall constitute failure of this test.

4.6.16 HERF. To verify conformance to 3.6.16, the pump assembly, in its operational configuration with fuel tank and pump housing filled with fuel, shall be tested IAW MIL-STD-464 for Hazards of Electromagnetic Radiation to Fuel. Failure to comply with MIL-STD-464 shall constitute failure of this test.

4.7 Personnel integration (MANPRINT). To verify conformance to 3.7, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Failure to provide the CoC in the FAT final report or comply with the requirements of 3.7 during FAT or CI shall constitute failure.

4.8 Identification, marking and information. To verify conformance to 3.8 and all subparagraphs, actual performance shall be recorded during the conduct of the FAT, and CoC shall be provided. Failure to provide the CoC in the FAT final report or comply with any requirement of 3.8 or any subparagraph during FAT or CI shall constitute failure.

4.9 Corrosion control performance. The requirements of 3.9 shall be demonstrated by engineering analysis, demonstrated applicable commercial/military experience and expertise on the same or similar items, and CoC. Absence of the CoC or failure to demonstrate compliance to the requirements of 3.9 shall constitute failure.

4.10 Treatment and painting. The requirements of 3.10 shall be demonstrated by color conformation, visual observation of workmanship and finish and CoC of materials and processes. Finish coat color shall be confirmed IAW 6.2, or the applicable delivery order in the contract. Absence of the CoC or failure to comply with any part of 3.10 or its referenced documents shall constitute failure.

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4.11 Welding. To verify conformance to 3.11, objective quality evidence shall be obtained, recorded and retained for Government review, and a CoC shall be provided. Separate CoCs shall be provided to verify that all material, workmanship, and documentation are in compliance with the applicable requirements of AWS D1.1/D1.1M, D1.2/D1.2M, and D1.3/D1.3M and the ASME Boiler and Pressure Vessel Code. Absence of any CoC or failure to demonstrate compliance to the requirements of 3.11 shall constitute failure.

4.12 NBC decontamination survivability. To verify conformance to 3.12, CoC shall be provided. Absence of the CoC or failure of the CoC to certify all the requirements of 3.12 shall constitute failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump assemblies are intended for transfer and dispensing of liquid fuel or potable water. The requirements are military unique since the commercial market does not meet the military requirements for NBC environment survivability, or have the durability requirements and mobility for off-road use.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this purchase description.
b. Type and Size of pump assembly required (see 1.2).
c. If first article inspection is not required. When required, the number of units required, and the time frame for submission (see 3.1, 4.1.1 and 6.3).
d. Spares packaging other than as specified (see 3.5.9.1).
e. If Triple Container (TRICON) is required, and Type required (see 3.5.10).
f. Identification and marking other than as specified (see 3.8).
g. Treatment and painting other than as specified (see 3.10)
h. Packaging required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a first article production unit(s). The first article should consist of one or more units. The contracting officer should include specific instructions in acquisitions documents regarding arrangements for examinations, approval of the first article examination results, and disposition of the first article(s).

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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals should be acquired under a separate contract line item in the contract.

6.5 Provisioning. The requirement for provisioning documentation for spare parts, consumables, and test measurement and diagnostic equipment should also be considered when this specification is applied on a contract.

6.6 Definitions.

6.6.1 Recovered materials. Recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials as distinguished from virgin raw materials.

6.6.2 Mission Orientated Protective Posture (MOPP). MOPP IV protection consists of a two piece protective overgarment, protective mask with hood, overboots and rubber gloves with cotton liners. The overgarment is closed and hood is pulled down and adjusted (Field manual FM-3-100).

6.6.3 Relevant hardware failure. A relevant hardware failure is any malfunction of contractor furnished hardware that:

a. Cannot be corrected within 15 minutes by the operator by adjustment, repair or replacement using on-board tools, spares, or repair parts furnished with the equipment; or
b. May cause failure to commence operation, cessation of operation, or degradation of performance more than 10% below specified levels; or
c. May damage assembly by continued operation; or
d. May cause a safety hazard to operating personnel.

6.6.4 Maintenance ratio. The maintenance ratio is the ratio of the total active maintenance in man-hours required (scheduled and unscheduled) to the total operating time.

6.6.5 Special tools and test equipment. Special tools and test equipment are defined as not found in the Army's General Mechanics tool kit (NSN 5180-00-177-7033), Organizational Maintenance Common #1 tool kit (NSN 4910-00-754-0654), Common #2 tool kit (NSN 4910-00-754-0650), and tool kit Supplement #1 (NSN 4910-00-754-0653), and U.S. Army Supply Catalogs 4910-95-A73 and 4910-95-A74. These kits and other tool kits/sets (US Army) are managed by USA TACOM-Rock Island, AMSTA-AC-CTTS, Rock Island, IL, 61299.

6.6.6 PMCS. PMCS includes assembly, disassembly, adjustments, maintenance, diagnose and repair or reporting of the condition of the assembly during the required system service life.

6.6.7 Operator's and occasionally occupied positions. The operator's position is defined as 24 inch horizontally and 12 inch vertically from the assembly control panel. Occasionally occupied positions are defined as anywhere within a 20-ft diameter around the assembly but outside a circle of 4-ft diameter around the engine and 55 inch above the ground.

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6.7 Highway transport. The 4-meter height restriction is based on AMOVP-1.

6.8 NATO intervehicle receptacle, and cable and plug assembly. Drawing 11674728 is provided as reference for a receptacle conforming to NATO STANAG 4074. Drawing 11682336-1 or NSN 6150-01-022-6004 is provided as reference for a cable and plug assembly conforming to NATO STANAG 4074.

6.9 NATO fuel designations. STANAG 1135 and AFLP 3747 describe the NATO fuel designations F-24, F-34, F-35, F-44 and F-54.

6.10 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for specification MIL-PRF-53051:
M53051-2-1
- 1 - Size I: 350 gpm capacity at 275 feet total head
- 2 - Type II: Pump Assembly, Water

6.11 Changes from previous issue. Marginal notations are used with a vertical line in this revision to identify changes with respect to the previous issue.

6.12 Subject term (key word) listing.

Centrifugal	350 GPM
Self-Priming	600 GPM
Liquid	Regulator
Aviation	Regulated
Diesel	Unregulated

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH

Preparing Activity:
Army – AT

(Project: 4320-2022-002)

Review Activities:
DLA – CC
Navy – YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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